AVL Trees, Red-Black Trees & Binary Search Tree Implementations

Complete notes covering BST insertion/deletion, AVL balance factor calculations and rotations, time complexity analysis, and practice coding problems.

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1. Binary Search Tree (BST) Properties

For every node X, left subtree keys are strictly less than X.key, and right subtree keys are strictly greater. Inorder traversal yields sorted sequence. In degenerate trees, search time degrades to O(N).

2. AVL Self-Balancing Trees

AVL trees maintain Balance Factor (height(left) - height(right)) in {-1, 0, +1}. Imbalances are resolved via 4 rotation types: Left-Left (Single Right), Right-Right (Single Left), Left-Right, and Right-Left.